DO YOU KNOW?

If $f(x) = \begin{cases} \alpha(x) & if \quad x < a \\ k & if \quad x = a \\ \beta(x) & if \quad x > a \end{cases}$, then

$L[f(x)]_{x=a} = \lim_{x \to a^-} f(x) = \lim_{x \to a} \alpha(x)$

$R[f(x)]_{x=a} = \lim_{x \to a^+} f(x) = \lim_{x \to a} \beta(x)$

It is applicable only when the function has different definition on both sides of the given point $x = a$.

Exercise 5.3

1. Examine the following functions for continuity at indicated points

(a) $f(x) = \begin{cases} \frac{x^2-4}{x-2}, & if \quad x \neq 2 \\ 0, & if \quad x = 2 \end{cases}$ at $x = 2$

(b) $f(x) = \begin{cases} \frac{x^2-9}{x-3}, & if \quad x \neq 3 \\ 6, & if \quad x = 3 \end{cases}$ at $x = 3$

2. Show that $f(x) = |x|$ is continuous at $x = 0$

5.2.10 Differentiability at a point

Let $f(x)$ be a real valued function defined on an open interval (a, b) and let $c \in (a, b)$. $f(x)$ is said to be differentiable or derivable at $x = c$ if and only if $\lim_{x \to c} \frac{f(x)-f(c)}{x-c}$ exists finitely.

This limit is called the derivative (or) differential co-efficient of the function $f(x)$ at $x = c$ and is denoted by $f'(c)$ (or) $D f(c)$ (or) $\left[\frac{d}{dx}(f(x))\right]_{x=c}$.

Thus $f'(c) = \lim_{x \to c} \frac{f(x)-f(c)}{x-c}$

5.2.11 Left hand derivative and right hand derivative

(i) $\lim_{x \to c^-} \frac{f(x)-f(c)}{x-c}$ or

$\lim_{h \to 0} \frac{f(c-h)-f(c)}{-h}$ is called the left hand derivative of $f(x)$ at $x = c$ and it denoted by $f'(c^-)$ or $L[f'(c)]$.

(ii) $\lim_{x \to c^+} \frac{f(x)-f(c)}{x-c}$ or

$\lim_{h \to 0} \frac{f(c+h)-f(c)}{h}$ is called the right hand derivative of $f(x)$ at $x = c$ and it denoted by $f'(c^+)$ or $R[f'(c)]$.

Result $f(x)$ is differentiable at $x = c \iff L[f'(c)] = R[f'(c)]$

Remarks:

(i) If $L[f'(c)] \neq R[f'(c)]$, then we say that $f(x)$ is not differentiable at $x=c$.

(ii) $f(x)$ is differentiable at $x = c \Rightarrow f(x)$ is continuous at $x = c$.

DO YOU KNOW?

A function may be continuous at a point but may not be differentiable at that point.

Example 5.29

Show that the function $f(x) = |x|$ is not differentiable at $x = 0$.

Solution

$f(x) = |x| = \begin{cases} x & if \quad x \geq 0 \\ -x & if \quad x < 0 \end{cases}$

$L[f'(0)] = \lim_{x \to 0^-} \frac{f(x)-f(0)}{x-0}$,

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